Thesis Plan

Student Details

Student name: Alex Gu
Student number: 48028903
Program enrolment: Bachelor of Computer Science (Honours)
Course enrolment: REIT4881
Number of units: 8
Expected workload for your enrolment (hours per week): 20

Supervisor Details

Primary Supervisor: Ruihong Qiu
Other Supervisors (optional): Zi (Helen) Huang
Have you met your supervisor this semester (yes/no, online/in person): Yes, in person
Do you have a regular meeting scheduled for this semester (weekly/biweekly/other—please specify): Other, meetings scheduled as needed

Thesis plan (thesis draft)

Use of Generative AI Statement

Have you used AI in your project

- Explore your topic? Yes – discussed federated learning and privacy concepts with Claude to build understanding
- Define your research question? No – research question was defined through meetings with supervisors
- Assist with your literature review? Yes – used Claude to generate notes, verified against originals
- Assist with formatting the answers presented here? Yes – used Claude for LaTeX troubleshooting and guidance on restructuring proposal material into the thesis template
- To generate figures? No – figures were created manually
- To analyse data? No – no experimental data has been analysed yet; analysis will be performed with my own code
- To generate data? No – all data comes from publicly available Kaggle datasets
- To generate text for your thesis? No – thesis text is my own writing, reused from my project proposal where permitted
- Other aspects of your work? Yes – feedback on writing voice and tone



THE UNIVERSITY OF QUEENSLAND
AUSTRALIA

Privacy-Preserving Federated Learning for Distributed Skin Disease Classification

by
ALEX GU

School of Electrical Engineering and Computer Science,
The University of Queensland.

Submitted for the degree of Bachelor of Computer Science (Honours) as part
of the course REIT4881.

August 2026

Alex Gu
alex.gu@student.uq.edu.au

August 2026

Prof Michael Bruenig
Head of School
School of Electrical Engineering and Computer Science
The University of Queensland
St Lucia, Q 4072

Dear Professor Bruenig,

In accordance with the requirements of the Degree of Bachelor of Computer Science (Honours) in the School of Electrical Engineering and Computer Science, I submit the following thesis entitled

"Privacy-Preserving Federated Learning for Distributed Skin Disease Classification"

The thesis was performed under the supervision of Dr Ruihong Qiu. I declare that the work submitted in the thesis is my own, except as acknowledged in the text and footnotes, and that it has not previously been submitted for a degree at the University of Queensland or any other institution.

Yours sincerely,

Alex Gu

ALEX G

Declaration by Author

This thesis is composed of my original work, and contains no material previously published, written by another person, or generated by artificial intelligence (AI), except where clearly referenced and acknowledged either in the main text or the preliminary pages of this thesis. I have clearly stated the contribution of others to jointly authored works that I have included in my thesis.

I have clearly stated the contribution of others to my thesis as a whole, including statistical assistance, survey design, data analysis, significant technical procedures, professional editorial advice, use of AI tools, financial support, and any other original research work used or reported in my thesis. The content of my thesis is the result of work I have carried out since the commencement of my enrollment in the course code indicated on the title page, and does not include a substantial part of work that has been submitted to qualify for the award of any other degree or diploma in any university or other tertiary institution. I have clearly stated which parts of my thesis, if any, have been submitted to qualify for another award.

I acknowledge that copyright of all material contained in my thesis resides with the copyright holder(s) of that material. Where appropriate I have obtained copyright permission from the copyright holder to reproduce material in this thesis and have sought permission from co-authors for any jointly authored works included in the thesis.

Use of AI Statement

I acknowledge the use of generative AI tools in completing this assessment. Details of which tools were used and how they were used are provided in the table below, along with appropriate in-text and full references. I take responsibility for critically evaluating and integrating the AI generated content, and ensuring it adheres to academic integrity standards.

AI model and date	**Artificial Intelligence (AI) tool usage** (tick all that apply)										
	Topic exploration	Research question	Literature review	Methodological support	Data analysis	Visual content creation	Written content creation	Language translation	Grammar/style/spelling	Feedback on content	Other use (provide details)
Claude Opus 4.6 09/03/26–17/03/26	✓		✓						✓	✓	LaTeX formatting and troubleshooting
Claude Opus 4.6 17/03/26–24/04/26									✓	✓	LaTeX formatting and troubleshooting
Claude Fable 5 23/04/26–12/08/26									✓	✓	LaTeX formatting; restructuring proposal into thesis template

Acknowledgments

I thank my supervisors, Ruihong Qiu and Zi (Helen) Huang, for their guidance on the direction and scope of this project. I also thank Laith Alzubaidi, who proposed the project and its staged structure, and whose advice has shaped the technical approach throughout.

Abstract

Skin disease classification using deep learning has shown strong potential but training accurate models requires large, diverse datasets often distributed across healthcare institutions. Privacy regulations such as the Health Insurance Portability and Accountability Act (HIPAA) and the General Data Protection Regulation (GDPR) prevent centralised data aggregation. Prior work has shown that even sharing model updates rather than raw data can leak sensitive patient information. Federated learning offers a framework for collaborative training across distributed nodes without centralising data, but standard federated averaging lacks convergence guarantees. This project evaluates federated learning approaches for multi-class skin disease classification in a simulated distributed healthcare environment. A working dataset of approximately 20,000 images across 30 diagnostic classes, drawn from publicly available skin disease image collections, will be partitioned across multiple nodes. FedAvg and FedProx will be implemented and compared against a centralised training baseline, evaluating classification accuracy, communication efficiency, and model behaviour under data that is not independently and identically distributed (non-IID). The findings of this project will provide guidance for selecting and configuring federated learning for medical image classification tasks in privacy-constrained distributed settings.

Contents

Declaration by Author iii

Use of AI Statement iv

Acknowledgments v

Abstract vi

List of Figures ix

List of Tables x

1 Introduction 1

2 Background and Literature review 3
- 2.1 Skin Disease Classification with Deep Learning . . . 3
- 2.2 Federated Learning . . . 3
- 2.3 Handling Non-IID Data in Federated Learning . . . 5
- 2.4 Privacy in Federated Learning . . . 5
- 2.5 Federated Learning in Medical Imaging . . . 6
- 2.6 Summary and Knowledge Gap . . . 6

3 Methodology 8
- 3.1 Project Aims . . . 8
- 3.2 Experimental Pipeline . . . 8
  - 3.2.1 Dataset . . . 8
  - 3.2.2 Centralised Baseline . . . 9
  - 3.2.3 Federated Learning . . . 9
  - 3.2.4 Evaluation . . . 10
- 3.3 Ethical Considerations . . . 10

4 Study 1: Centralised Baseline and IID Federated Learning 12
- 4.1 Aims . . . 12
- 4.2 Methods . . . 12
- 4.3 Results . . . 12

4.4 Discussion 12

5 Study 2: Federated Learning under Non-IID Conditions 13
5.1 Aims 13
5.2 Methods 13
5.3 Results 13
5.4 Discussion 13

6 General Discussion and Conclusions 14
6.1 Summary 14
6.2 Discussion and Future Work 14
6.3 Conclusions 14

Bibliography 15

Appendices 18

A Program listings 19

List of Figures

2.1 Overview of the federated learning pipeline 4

List of Tables

3.1 Summary statistics of the datasets used in this project. 9

Chapter 1

Introduction

Skin diseases are among the most common health conditions worldwide [1] and deep learning has demonstrated strong performance in skin disease classification from images [2]. However, training models that generalise well requires diverse data across patient demographics and imaging setups. In practice this data sits in separate hospitals and research groups, each with a relatively small local dataset. Privacy regulations including the Health Insurance Portability and Accountability Act (HIPAA) and the General Data Protection Regulation (GDPR) restrict the centralisation of patient data [3], preventing the ability to build robust classifiers through conventional centralised training.

Federated learning addresses this by enabling multiple institutions to collaboratively train a shared model without exchanging raw data. Each participant trains on its local dataset and shares only model updates with a central server, which aggregates them into a global model. McMahan et al. introduced Federated Averaging (FedAvg), which reduces communication costs by performing multiple local training epochs before aggregation [4]. However, FedAvg lacks convergence guarantees when data across participants is not independently and identically distributed (non-IID), a condition that is common in healthcare settings where different institutions may see different patient demographics and disease distributions. FedProx extends FedAvg by introducing a proximal term that constrains local model updates, providing convergence guarantees under heterogeneous conditions [5]. Furthermore, federated learning alone does not fully resolve privacy concerns. Prior work has shown that shared model updates can leak sensitive information, and additional mechanisms may be needed [3].

This project evaluates federated learning for multi-class skin disease classification in a simulated distributed healthcare environment. A working dataset of approximately 20,000 images across 30 diagnostic classes, drawn from publicly available skin disease image collections, will be partitioned across multiple nodes. FedAvg and FedProx will be implemented and compared against a centralised training baseline, evaluating classification accuracy and communication efficiency under non-IID data distributions.

The remainder of this thesis is organised as follows. Chapter 2 reviews the relevant

literature on federated learning, data heterogeneity, and privacy. Chapter 3 describes the methodology, including the dataset, centralised baseline, and federated learning configurations. Chapters 4 and 5 present the experimental studies, and Chapter 6 concludes.

Chapter 2

Background and Literature review

2.1 Skin Disease Classification with Deep Learning

Skin diseases are among the most prevalent health conditions globally, affecting hundreds of millions of people and contributing significantly to the global burden of disease [1]. Accurate and timely diagnosis is critical for effective treatment, but access to dermatological expertise is limited in many regions. This has motivated research into automated classification systems that can assist clinicians by analysing images of skin conditions.

Deep learning, particularly convolutional neural networks (CNNs), has demonstrated strong performance in this domain. Esteva et al. [2] showed that a CNN trained on a large dataset of clinical images could classify skin cancer at a level comparable to board-certified dermatologists, establishing deep learning as a viable tool for skin disease diagnosis. Since then, the field has benefited from the availability of large public datasets such as HAM10000, a collection of 10,015 dermatoscopic images spanning seven diagnostic categories [6]. These datasets have enabled accelerated progress in automated skin disease classification.

The remaining bottleneck is data access. Generalisation across skin tones and imaging hardware requires broader datasets than any single institution holds. In practice, such data is distributed across hospitals and associated health and research institutions, and privacy regulations prevent its centralisation.

2.2 Federated Learning

Federated learning offers a solution for training machine learning models across distributed data sources without centralising the data itself. McMahan et al. [4] introduced Federated Averaging (FedAvg) which frames the learning problem as minimising a global objective function across K clients:

$$\min_{w} f(w) = \sum_{k=1}^{K} \frac{n_k}{n} F_k(w) \tag{2.1}$$

where w are the model parameters, $F_k(w)$ is the local objective function for client k, n_k is the number of data points held by client k, and n is the total number of data points across all clients. Each client's contribution to the global objective is weighted by the proportion of data it holds. Figure 2.1 illustrates this process.

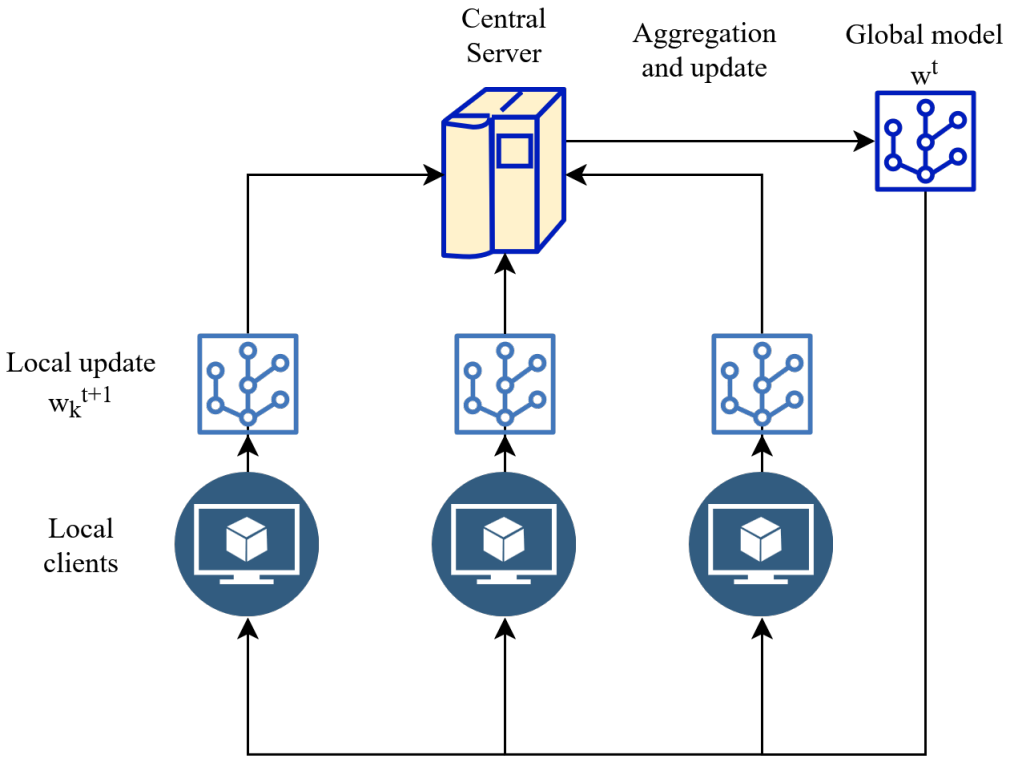


Figure 2.1: Overview of the federated learning pipeline

In each communication round, the server distributes the current global model w^t to a subset of clients. Each selected client initialises its local model with w^t and performs E epochs of stochastic gradient descent on its local data, producing updated parameters w_k^{t+1}. The server then aggregates these updates by computing a weighted average:

$$w^{t+1} = \sum_{k=1}^{K} \frac{n_k}{n} w_k^{t+1} \tag{2.2}$$

By performing multiple local training epochs before communication, FedAvg significantly reduces communication costs compared to approaches that require gradient updates after every training step.

Li et al. [7] surveyed the main challenges. 1) Communication costs are high when bandwidth is limited or models are large. 2) Participating devices vary in compute and connectivity. 3) Data across nodes is usually non-IID. 4) Shared updates can still leak information about training data.

2.3 Handling Non-IID Data in Federated Learning

Data distributions across participating nodes are rarely uniform. When each node holds data that is non-IID, local models trained on these subsets may diverge significantly from one another. Averaging such divergent models can produce a global model that performs poorly compared to one trained on centralised data.

FedAvg is particularly susceptible to this problem as it provides no way to limit how far local models can deviate from the global model during local training. Li et al. [5] proposed FedProx to address this limitation. FedProx modifies the local objective function by adding a proximal term:

$$\min_{w} h_k(w; w^t) = F_k(w) + \frac{\mu}{2}|w - w^t|^2 \tag{2.3}$$

where $F_k(w)$ is the local loss, w^t is the current global model and μ controls the strength of the penalty. This term penalises large deviations from the current global model. This modification provides convergence guarantees under heterogeneous data distributions, which FedAvg lacks. FedProx also accommodates systems heterogeneity by allowing nodes to perform variable amounts of local computation.

Other approaches have been proposed to tackle non-IID issues. Karimireddy et al. [8] introduced SCAFFOLD, which uses control variates to correct for the drift in local updates. This reduces the variance caused by heterogeneous data. Wang et al. [9] proposed Federated Matched Averaging (FedMA), which aligns neurons across local models before averaging. This prevents neurons in different local models from learning equivalent features in different positions. Wang et al. [10] proposed FedNova, which normalises local updates to account for differences in the number of local training steps, correcting an inconsistency in FedAvg that is amplified under heterogeneous conditions. While this project focuses on FedAvg and FedProx, these alternative approaches provide context for understanding the breadth of strategies available for handling non-IID data in federated settings.

To evaluate federated learning methods under heterogeneous conditions, a consistent experimental method is required for simulating non-IID data splits. A widely adopted approach, introduced by Hsu et al. [11], is to partition data across clients using a Dirichlet distribution over class labels, parameterised by a concentration parameter α. Lower values of α produce more skewed label distributions across clients while higher values approach IID. This provides a controllable and reproducible way to study model behaviour across a wide range of heterogeneous conditions.

2.4 Privacy in Federated Learning

Federated learning avoids sharing raw data but this alone does not guarantee privacy. Kaissis et al. [3] noted that the model updates transmitted during training can be exploited to infer information about the underlying data. Model inversion attacks use model outputs

or gradients to reconstruct approximations of training inputs [12]. Membership inference attacks determine whether a specific data point was in the training set by analysing the model's response to that input [13].

Several defence mechanisms have been proposed to strengthen privacy in federated learning. Bonawitz et al. [14] designed a practical secure aggregation protocol using pairwise masking to protect each participant's contribution. This enables the server to compute the sum of client updates without observing any individual update. This protocol is robust to client dropout, making it suitable for real-world deployments. Differential privacy provides a complementary approach by adding calibrated noise to the data or model updates to limit the information that can be inferred about any individual training example. Dwork et al. [15] established the formal foundations of differential privacy, while Abadi et al. [16] introduced DP-SGD, a training procedure that clips per-example gradients and adds noise during optimisation which enables deep learning models to be trained with formal privacy guarantees. These mechanisms can be combined with federated learning to provide stronger privacy protections, though they often introduce trade-offs with model accuracy.

2.5 Federated Learning in Medical Imaging

Federated learning has attracted significant interest in medical imaging, where data is inherently distributed across institutions and subject to strict privacy requirements. Rieke et al. [17] outlined the potential of federated learning to enable collaborative model development across hospitals without centralising patient data, identifying it as a key enabler for the future of digital health. They noted that the heterogeneity of clinical data, arising from differences in imaging equipment, protocols, and patient demographics makes the non-IID problem particularly important in this domain.

Sheller et al. [18] demonstrated that federated learning could be applied to brain tumour segmentation across multiple institutions. They achieved performance comparable to centralised training while keeping patient data local. Their work provided early evidence that federated learning is viable in medical imaging tasks with complex, heterogeneous data. More recently, Dayan et al. [19] conducted one of the largest federated learning studies to date, training a model to predict clinical outcomes in COVID-19 patients across 20 institutions worldwide. Their model outperformed any single-institution model, which justifies federated learning in practice.

2.6 Summary and Knowledge Gap

The preceding sections have reviewed the relevant literature across skin disease classification, federated learning, non-IID data handling, privacy mechanisms, and federated learning in medical imaging. Several key gaps emerge from this review:

1. **Limited evaluation of federated learning for multi-class skin disease classi-**

fication. Federated learning has been applied successfully in medical imaging, but primarily in radiology and pathology tasks with binary or few-class classification. Its performance on fine-grained, multi-class skin disease tasks with 20 or more classes has not been systematically studied.

2. **Insufficient comparison of federated algorithms under realistic non-IID conditions.** While FedProx was designed to address the non-IID limitations of FedAvg, direct comparisons between the two on medical imaging tasks with realistic heterogeneity remain scarce. Most evaluations use standard benchmarks rather than domain-specific datasets where class imbalance and label skew occur.

3. **Lack of practical guidance on federated learning trade-offs for clinical imaging.** Existing studies report accuracy but often do not systematically evaluate communication efficiency or per-class performance alongside non-IID sensitivity. Understanding these trade-offs is essential for informing whether federated learning is practical for real-world deployment.

This project addresses these gaps by evaluating FedAvg and FedProx on a 30-class skin disease dataset to simulate realistic data heterogeneity. By comparing these approaches against a centralised baseline on classification accuracy, communication efficiency, and behaviour under non-IID distributions, this project aims to provide practical insight into the suitability of federated learning for fine-grained medical image classification.

Chapter 3

Methodology

3.1 Project Aims

Based on the knowledge gaps identified in the preceding chapter, this project has the following aims:

- Train a centralised ResNet classifier on 20,000 images across 30 classes to establish an upper-bound benchmark
- Run FedAvg and FedProx on the same data, partitioned into IID and Dirichlet non-IID splits across simulated clients
- Report how each federated method degrades from the centralised baseline as heterogeneity increases, and what it costs in communication rounds

3.2 Experimental Pipeline

This project follows a three-stage experimental pipeline: centralised baseline training, federated learning implementation, and comparative evaluation. All experiments will be implemented in Python using PyTorch.

3.2.1 Dataset

Two publicly available skin disease image datasets from Kaggle will be used. The first, the Skin Diseases and Cancer Comprehensive Dataset, contains approximately 13,000 images across 24 diagnostic classes [20]. The second, a larger balanced skin disease dataset, contains approximately 262,000 images across 34 classes [21]. Table 3.1 summarises the two datasets. The final dataset composition will be determined during implementation; options include using either dataset independently, merging them with a unified class mapping, or using a subsampled combination to manage compute requirements. Images will be resized to a uniform resolution and normalised to standardise input across all

experiments. Standard data augmentation techniques such as random horizontal flipping, rotation, and colour jittering will be applied during training to improve generalisation. The dataset will be split into training, validation, and test sets, with the test set held centrally and kept consistent across all experiments to ensure fair comparison.

Table 3.1: Summary statistics of the datasets used in this project.

Property	**Dataset 1**	**Dataset 2**
Name	Skin Diseases + Cancer	Massive Skin Disease
Source	Kaggle	Kaggle
Total images	∼13,000	∼262,000
Number of classes	24	34
Image type	Clinical photographs	Clinical photographs
Class balance	Imbalanced	Balanced

3.2.2 Centralised Baseline

A ResNet-18 architecture pre-trained on ImageNet will be used as the primary classification model across all experiments, selected for its established performance in image classification tasks [22]. If compute resources permit, ResNet-50 will also be evaluated to assess the impact of model capacity on classification performance. The baseline will be trained on the full training set in a standard centralised setting, with all data aggregated on a single machine. This represents an upper bound on classification performance. Training will use stochastic gradient descent with momentum, with learning rate, batch size, and number of epochs tuned on the validation set. To account for variability, experiments will be repeated across multiple random seeds and results reported as mean and standard deviation.

3.2.3 Federated Learning

The training set will be partitioned across K simulated client nodes to replicate a distributed healthcare environment. Two partitioning strategies will be used. In the IID setting, data will be distributed uniformly at random across clients, ensuring each client holds a roughly representative subset. In the non-IID setting, a Dirichlet distribution with concentration parameter α will be used to create heterogeneous label distributions across clients, where lower values of α produce greater skew [11]. Multiple values of α (for example, $\alpha \in \{0.1, 0.5, 1.0\}$) will be tested to evaluate model behaviour under varying degrees of heterogeneity. The inherent class imbalance in the dataset will be considered when designing the partitioning, as it interacts with the label skew introduced by non-IID splits.

FedAvg and FedProx will each be implemented following the procedures described in Chapter 2. The same ResNet-18 architecture and preprocessing pipeline used in the

centralised baseline will be applied to ensure that differences in performance are attributable to the training method rather than the model.

In each communication round, a fraction $C = 0.5$ of $K = 10$ clients will be selected to participate. Each selected client will train on its local data for $E = 5$ local epochs with batch size $B = 32$, then send its updated model parameters to the central server for aggregation.

For FedProx, the proximal term weight μ will be tuned over $\mu \in \{0.001, 0.01, 0.1, 1.0\}$ using the validation set. This allows assessment of its impact under different levels of regularisation.

Training will run for up to $T = 100$ communication rounds. Early stopping will be applied based on validation performance.

3.2.4 Evaluation

All approaches will be evaluated on the same held-out test set using the following metrics.

- **Classification accuracy**: overall accuracy on the test set.
- **Per-class performance**: precision, recall, and F1-score per class to assess behaviour on minority classes, which is particularly important given dataset imbalance.
- **Communication efficiency**: the number of communication rounds required to reach a target accuracy and the total volume of parameter data transmitted between clients and server.
- **Non-IID sensitivity**: accuracy as a function of the Dirichlet concentration parameter α, measuring how each method degrades under increasing data heterogeneity.

Results from FedAvg and FedProx will be compared against each other and against the centralised baseline across all metrics. All experiments will be repeated over multiple random seeds to ensure reported differences are statistically meaningful.

3.3 Ethical Considerations

This project uses only publicly available skin disease image datasets obtained from Kaggle. Both datasets have been released by their authors for research and educational purposes, and contain no directly identifying patient information such as names or dates of birth. No new patient data will be collected, and no human or animal subjects will be involved in the research. For these reasons, formal ethics approval is not required for this project.

Several ethical considerations remain relevant. Publicly released medical image datasets may still carry risks of re-identification. This project will treat all data as sensitive and will not attempt to link images back to individuals or external sources.

Skin disease classifiers are known to exhibit performance disparities across skin tones, as publicly available datasets are often skewed toward lighter skin types. Any results produced by this project will be interpreted with this limitation in mind and per-class performance metrics will be reported to surface any systematic disparities that arise.

While this project is a simulation and does not involve real clinical deployment, the broader motivation carries downstream implications for patient privacy, model accountability, and the trustworthiness of clinical AI systems. This project is a methodological evaluation and no diagnostic claims follow from it.

All code, experimental configurations, and results will be managed with version control to support reproducibility, and the datasets used will be properly cited so that others can verify and build upon the work.

Chapter 4

Study 1: Centralised Baseline and IID Federated Learning

4.1 Aims

- Establish centralised ResNet-18 baseline performance on the 30-class dataset.
- Evaluate FedAvg and FedProx under IID partitioning to verify that federated training approaches the centralised upper bound when data is homogeneous.

4.2 Methods

- Centralised training as described in Chapter 3, with hyperparameters tuned on the validation set.
- FedAvg and FedProx with $K = 10$ clients, $C = 0.5$, $E = 5$, $B = 32$, uniform random data partitioning.

4.3 Results

- To be completed: accuracy and per-class metrics for the centralised baseline.
- To be completed: federated accuracy versus communication rounds under IID conditions.

4.4 Discussion

- To be completed: gap between federated and centralised performance under IID conditions.

Chapter 5

Study 2: Federated Learning under Non-IID Conditions

5.1 Aims

- Evaluate how FedAvg and FedProx degrade as data heterogeneity increases.
- Assess whether the FedProx proximal term improves stability and accuracy under label skew.

5.2 Methods

- Dirichlet partitioning with $\alpha \in \{0.1, 0.5, 1.0\}$.
- FedProx proximal weight tuned over $\mu \in \{0.001, 0.01, 0.1, 1.0\}$.

5.3 Results

- To be completed: accuracy as a function of α for both methods.
- To be completed: communication rounds to target accuracy, per-class performance on minority classes.

5.4 Discussion

- To be completed: comparison of FedAvg and FedProx degradation, practical implications for federated medical imaging.

Chapter 6

General Discussion and Conclusions

6.1 Summary

- To be completed: summary of baseline, IID, and non-IID findings.

6.2 Discussion and Future Work

- To be completed: how the two studies together answer the project aims.
- Planned future directions: stronger architectures such as Vision Transformers, non-IID improvements, communication optimisations.

6.3 Conclusions

- To be completed: overall assessment of whether federated learning is a viable substitute for centralised training in this setting, and under what heterogeneity conditions.

Bibliography

[1] Roderick J. Hay, Nicole E. Johns, Hywel C. Williams, Ian W. Bolliger, Robert P. Dellavalle, David J. Margolis, Robin Marks, Luigi Naldi, Martin A. Weinstock, Sarah K. Wulf, Catherine Michaud, Christopher J L Murray, and Mohsen Naghavi. The global burden of skin disease in 2010: An analysis of the prevalence and impact of skin conditions. *The Journal of Investigative Dermatology*, 134(6):1527–1534, June 2014. ISSN 1523-1747. 10.1038/jid.2013.446.

[2] Andre Esteva, Brett Kuprel, Roberto A. Novoa, Justin Ko, Susan M. Swetter, Helen M. Blau, and Sebastian Thrun. Dermatologist-level classification of skin cancer with deep neural networks. *Nature*, 542(7639):115–118, February 2017. ISSN 1476-4687. 10.1038/nature21056.

[3] Georgios A. Kaissis, Marcus R. Makowski, Daniel Rückert, and Rickmer F. Braren. Secure, privacy-preserving and federated machine learning in medical imaging. *Nature Machine Intelligence*, 2(6):305–311, June 2020. ISSN 2522-5839. 10.1038/s42256-020-0186-1.

[4] H. Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas. Communication-Efficient Learning of Deep Networks from Decentralized Data, January 2023.

[5] Tian Li, Anit Kumar Sahu, Manzil Zaheer, Maziar Sanjabi, Ameet Talwalkar, and Virginia Smith. Federated Optimization in Heterogeneous Networks, April 2020.

[6] Philipp Tschandl, Cliff Rosendahl, and Harald Kittler. The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions. *Scientific Data*, 5(1):180161, August 2018. ISSN 2052-4463. 10.1038/sdata.2018.161.

[7] Tian Li, Anit Kumar Sahu, Ameet Talwalkar, and Virginia Smith. Federated Learning: Challenges, Methods, and Future Directions. *IEEE Signal Processing Magazine*, 37 (3):50–60, May 2020. ISSN 1558-0792. 10.1109/MSP.2020.2975749.

[8] Sai Praneeth Karimireddy, Satyen Kale, Mehryar Mohri, Sashank J. Reddi, Sebastian U. Stich, and Ananda Theertha Suresh. SCAFFOLD: Stochastic Controlled Averaging for Federated Learning, April 2021.

[9] Hongyi Wang, Mikhail Yurochkin, Yuekai Sun, Dimitris Papailiopoulos, and Yasaman Khazaeni. Federated Learning with Matched Averaging, February 2020.

[10] Jianyu Wang, Qinghua Liu, Hao Liang, Gauri Joshi, and H. Vincent Poor. Tackling the Objective Inconsistency Problem in Heterogeneous Federated Optimization, July 2020.

[11] Tzu-Ming Harry Hsu, Hang Qi, and Matthew Brown. Measuring the Effects of Non-Identical Data Distribution for Federated Visual Classification, September 2019.

[12] Matt Fredrikson, Somesh Jha, and Thomas Ristenpart. Model Inversion Attacks that Exploit Confidence Information and Basic Countermeasures. In *Proceedings of the 22nd ACM SIGSAC Conference on Computer and Communications Security*, CCS '15, pages 1322–1333, New York, NY, USA, October 2015. Association for Computing Machinery. ISBN 978-1-4503-3832-5. 10.1145/2810103.2813677.

[13] Reza Shokri, Marco Stronati, Congzheng Song, and Vitaly Shmatikov. Membership Inference Attacks Against Machine Learning Models. In *2017 IEEE Symposium on Security and Privacy (SP)*, pages 3–18, May 2017. 10.1109/SP.2017.41.

[14] Keith Bonawitz, Vladimir Ivanov, Ben Kreuter, Antonio Marcedone, H. Brendan McMahan, Sarvar Patel, Daniel Ramage, Aaron Segal, and Karn Seth. Practical Secure Aggregation for Privacy-Preserving Machine Learning. In *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security*, CCS '17, pages 1175–1191, New York, NY, USA, October 2017. Association for Computing Machinery. ISBN 978-1-4503-4946-8. 10.1145/3133956.3133982.

[15] Cynthia Dwork, Frank McSherry, Kobbi Nissim, and Adam Smith. Calibrating Noise to Sensitivity in Private Data Analysis. In Shai Halevi and Tal Rabin, editors, *Theory of Cryptography*, pages 265–284, Berlin, Heidelberg, 2006. Springer. ISBN 978-3-540-32732-5. 10.1007/11681878_14.

[16] Martín Abadi, Andy Chu, Ian Goodfellow, H. Brendan McMahan, Ilya Mironov, Kunal Talwar, and Li Zhang. Deep Learning with Differential Privacy. In *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security*, pages 308–318, October 2016. 10.1145/2976749.2978318.

[17] Nicola Rieke, Jonny Hancox, Wenqi Li, Fausto Milletarì, Holger R. Roth, Shadi Albarqouni, Spyridon Bakas, Mathieu N. Galtier, Bennett A. Landman, Klaus Maier-Hein, Sébastien Ourselin, Micah Sheller, Ronald M. Summers, Andrew Trask, Daguang Xu, Maximilian Baust, and M. Jorge Cardoso. The future of digital health with federated learning. *npj Digital Medicine*, 3(1):119, September 2020. ISSN 2398-6352. 10.1038/s41746-020-00323-1.

[18] Micah J. Sheller, Brandon Edwards, G. Anthony Reina, Jason Martin, Sarthak Pati, Aikaterini Kotrotsou, Mikhail Milchenko, Weilin Xu, Daniel Marcus, Rivka R. Colen, and Spyridon Bakas. Federated learning in medicine: Facilitating multi-institutional collaborations without sharing patient data. *Scientific Reports*, 10(1):12598, July 2020. ISSN 2045-2322. 10.1038/s41598-020-69250-1.

[19] Ittai Dayan, Holger R. Roth, Aoxiao Zhong, Ahmed Harouni, Amilcare Gentili, Anas Z. Abidin, Andrew Liu, Anthony Beardsworth Costa, Bradford J. Wood, Chien-Sung Tsai, Chih-Hung Wang, Chun-Nan Hsu, C. K. Lee, Peiying Ruan, Daguang Xu, Dufan Wu, Eddie Huang, Felipe Campos Kitamura, Griffin Lacey, Gustavo César de Antônio Corradi, Gustavo Nino, Hao-Hsin Shin, Hirofumi Obinata, Hui Ren, Jason C. Crane, Jesse Tetreault, Jiahui Guan, John W. Garrett, Joshua D. Kaggie, Jung Gil Park, Keith Dreyer, Krishna Juluru, Kristopher Kersten, Marcio Aloisio Bezerra Cavalcanti Rockenbach, Marius George Linguraru, Masoom A. Haider, Meena AbdelMaseeh, Nicola Rieke, Pablo F. Damasceno, Pedro Mario Cruz e Silva, Pochuan Wang, Sheng Xu, Shuichi Kawano, Sira Sriswasdi, Soo Young Park, Thomas M. Grist, Varun Buch, Watsamon Jantarabenjakul, Weichung Wang, Won Young Tak, Xiang Li, Xihong Lin, Young Joon Kwon, Abood Quraini, Andrew Feng, Andrew N. Priest, Baris Turkbey, Benjamin Glicksberg, Bernardo Bizzo, Byung Seok Kim, Carlos Tor-Díez, Chia-Cheng Lee, Chia-Jung Hsu, Chin Lin, Chiu-Ling Lai, Christopher P. Hess, Colin Compas, Deepeksha Bhatia, Eric K. Oermann, Evan Leibovitz, Hisashi Sasaki, Hitoshi Mori, Isaac Yang, Jae Ho Sohn, Krishna Nand Keshava Murthy, Li-Chen Fu, Matheus Ribeiro Furtado de Mendonça, Mike Fralick, Min Kyu Kang, Mohammad Adil, Natalie Gangai, Peerapon Vateekul, Pierre Elnajjar, Sarah Hickman, Sharmila Majumdar, Shelley L. McLeod, Sheridan Reed, Stefan Gräf, Stephanie Harmon, Tatsuya Kodama, Thanyawee Puthanakit, Tony Mazzulli, Vitor Lima de Lavor, Yothin Rakvongthai, Yu Rim Lee, Yuhong Wen, Fiona J. Gilbert, Mona G. Flores, and Quanzheng Li. Federated learning for predicting clinical outcomes in patients with COVID-19. *Nature Medicine*, 27(10):1735–1743, October 2021. ISSN 1546-170X. 10.1038/s41591-021-01506-3.

[20] Aniete Orudofia. Skin Diseases + Cancer Comprehensive Dataset. `https://www.kaggle.com/datasets/anieetorudofia/skin-diseases-cancer-comprehensive-dataset`, n.d.

[21] Muhammad Abdul Sami. Massive skin disease balanced dataset. `https://www.kaggle.com/datasets/muhammadabdulsami/massive-skin-disease-balanced-dataset`, n.d.

[22] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep Residual Learning for Image Recognition, December 2015.

Appendix A

Program listings

Code and experimental configurations will be made available via a GitHub repository, with directions provided in the final thesis.